

Assignment 2

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0.1 Instructions

This problem set is based on Lecture 7-11. The questions are updated as we cover each lecture.

For each question:

1. Understand the economic intuition before using formulas.
2. Be clear about what the model is trying to answer.
3. Use concise reasoning.

1 Lecture 7: Interest-Rate Models

1.1 Question 1

A junior analyst says:

“Interest rates are uncertain, so we should use a short-rate model for every bond we price.”

The analyst is comparing the following securities:

Security	Description
A	5-year Treasury note with fixed annual coupons
B	5-year fixed-rate corporate bond with no embedded options
C	5-year callable agency bond
D	Interest-rate cap
E	Mortgage-backed security

Answer the following:

- a. Explain why Securities A and B usually do not require a stochastic short-rate model for time-0 valuation.
- b. State the basic valuation logic for a plain fixed-cash-flow bond using discount factors.
- c. Explain what a bootstrapped discount curve provides.
- d. Identify which securities in the table require future interest-rate dynamics and explain why.
- e. Correct the analyst’s statement in one or two sentences.

1.2 Question 2

A portfolio manager asks why Lecture 7 spends time modeling the short rate if today's yield curve is already observable.

Answer the following:

- a. Explain the difference between a yield curve and a short-rate model.
- b. Explain what is meant by saying that a yield curve is a “snapshot.”
- c. Explain what is meant by saying that a short-rate model is a “scenario generator.”
- d. Explain why future discount curves $P(t, T)$ matter for securities whose cash flows depend on future rates.
- e. Give two examples of portfolio or risk-management questions that cannot be answered well using only today's yield curve.

1.3 Question 3

Consider the basic one-factor short-rate model:

$$dr_t = b(r_t, t)dt + \sigma(r_t, t)dz_t.$$

Answer the following:

- a. Explain the role of the drift term.
- b. Explain the role of the volatility term.
- c. Explain what mean reversion means and why it may be a reasonable assumption for interest rates.
- d. Compare the intuition behind constant volatility, level-proportional volatility, and square-root volatility.
- e. Explain one practical consequence of choosing a normal model that allows negative rates.

1.4 Question 4

A fixed-income team is deciding between an equilibrium model and an arbitrage-free model.

Answer the following:

- Explain the main purpose of an equilibrium interest-rate model.
- Explain the main purpose of an arbitrage-free interest-rate model.
- Explain why arbitrage-free models are usually preferred for pricing derivatives and bonds with embedded options.
- In the Hull-White model,

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t,$$

explain why $\theta(t)$ is central to fitting today's curve.

- Explain why a and σ are usually calibrated to interest-rate option prices rather than plain bond prices.

1.5 Question 5

An analyst wants to estimate historical short-rate volatility from weekly data. The analyst observes the following weekly changes in the short rate:

Week	Change in short rate
1	+4 bps
2	-2 bps
3	+6 bps
4	-5 bps
5	+3 bps
6	+1 bp
7	-4 bps
8	+5 bps

Answer the following:

- Explain why rate changes, rather than rate levels, are used to estimate volatility in this setting.

- b. Describe the steps for estimating weekly historical volatility from these observations.
- c. Compute the sample standard deviation of the weekly changes in basis points.
- d. Annualize the weekly volatility using $\sqrt{52}$.
- e. Explain two limitations of using this historical volatility estimate as an input to a short-rate model.

2 Lecture 8: Bonds with Embedded Options and Option-Adjusted Measures

2.1 Question 6

A 3-year annual-pay corporate bond has face value 100 and coupon rate 5%. The Treasury zero rates are:

Maturity	Zero rate
1 year	3.00%
2 years	4.00%
3 years	5.00%

The bond trades at 99.00.

Answer the following:

- a. Write the promised cash flows of the bond.
- b. Compute the value of the bond if its promised cash flows were discounted using only the Treasury zero curve.
- c. Explain why the market price can be lower than this Treasury-discounted value.
- d. Set up the static spread equation for this bond.
- e. Use the following trial present values to estimate the static spread and explain why the price-spread relationship is nonlinear:

Spread	Present value
0 bps	100.18
50 bps	98.83
100 bps	97.50

2.2 Question 7

A 3-year annual-pay bond has face value 100 and coupon 7. The risk-neutral probability is $p = 0.5$. The short-rate lattice is:

State	Year 0	Year 1	Year 2
Low-rate state	5%	4%	3%
Middle state		8%	6%
High-rate state			10%

The bond is callable at 101 beginning in year 2.

Answer the following:

- Compute the terminal payoff at year 3.
- Compute the year-2 option-free hold values at the 3%, 6%, and 10% nodes.
- Apply the call rule at year 2. Which nodes are called?
- Roll the callable values back to year 1.
- Compute the callable bond value at time 0 and compare it with the option-free value.

2.3 Question 8

A 3-year annual-pay bond has face value 100 and coupon 4. The risk-neutral probability is $p = 0.5$. The short-rate lattice is:

State	Year 0	Year 1	Year 2
Low-rate state	5%	3%	4%
Middle state		9%	8%
High-rate state			12%

The bond is puttable at 100 beginning in year 1.

Answer the following:

- a. Compute the terminal payoff at year 3.
- b. Compute the year-2 option-free hold values.
- c. Apply the put rule at year 2. Which nodes are put?
- d. Roll the puttable values back to year 1 and apply the put rule again.
- e. Compute the puttable bond value at time 0 and compare it with the option-free value.

2.4 Question 9

An analyst says:

“A binomial interest-rate tree is determined by the starting short rate, volatility, and time step, so it cannot also fit the full zero curve.”

Assume a model uses the following structure:

$$r_{t,j} = a_t e^{(2j-t)\sigma}.$$

Answer the following:

- a. Explain what σ controls in this tree.
- b. Explain what the time-dependent level parameter a_t controls.
- c. Explain why a valuation-grade tree needs one calibration condition for each maturity on the zero curve.
- d. Suppose the zero curve has four annual maturities. How many level parameters are needed to fit the four discount factors?
- e. Correct the analyst’s statement in one or two sentences.

2.5 Question 10

A callable bond is priced with a calibrated 4-period lattice. The model price with OAS equal to 0 is 103.25, while the observed market price is 101.00. The analyst then tries several OAS values:

Trial OAS	Model price
0 bps	103.25
50 bps	101.80
75 bps	101.08
80 bps	100.94

After solving for OAS, the analyst computes effective duration and convexity using a 50 bp parallel curve shock. The model produces:

Scenario	Repriced value
Base curve	101.00
Curve down 50 bps	103.20
Curve up 50 bps	98.90

Answer the following:

- Explain what OAS is adding to the lattice.
- Estimate the OAS from the trial values.
- Explain why the OAS is positive in this example.
- Compute effective duration.
- Compute effective convexity and explain what these measures capture for a callable bond.

3 Lecture 9: Interest-Rate Futures Contracts

3.1 Question 11

A trader buys one Treasury futures contract at 100-08. The contract size is \$100,000, and one full point is worth \$1,000. The next three daily settlement prices are:

Day	Settlement price
0	100-08
1	100-20
2	99-28
3	100-04

Answer the following:

- Explain what it means to be long a futures contract.
- Convert each settlement price into decimal form.
- Compute the daily mark-to-market cash flow to the long position.
- Compute the cumulative cash flow over the three days.
- Explain why the trader does not pay the full notional value when entering the futures position.

3.2 Question 12

A trader buys one Three-Month SOFR futures contract at 95.36. The next settlement price is 95.42. Each 1-basis-point price move is worth approximately \$25 per contract.

Answer the following:

- Compute the initial implied SOFR rate.
- Compute the new implied SOFR rate.
- Compute the price change in basis points.
- Compute the dollar gain or loss for the long futures position.
- Explain why a higher SOFR futures price corresponds to a lower implied SOFR rate.

3.3 Question 13

A Treasury futures contract has settlement price 96.00. Three eligible deliverable bonds have the following cash prices and conversion factors:

Bond	Cash price	Conversion factor
A	101.40	1.0400
B	98.20	1.0150
C	94.10	0.9700

Ignore accrued interest and financing.

Answer the following:

- Explain what the cheapest-to-deliver bond is.
- Compute the invoice amount before accrued interest for each bond.
- Compute the net delivery cost for each bond.
- Identify the cheapest-to-deliver bond.
- Explain why the lowest cash price alone does not necessarily identify the cheapest-to-deliver bond.

3.4 Question 14

A bond sells for 100, pays a 5 percent annual coupon, and the futures contract settles in six months. The financing rate is 7 percent per year.

Answer the following:

- Compute the theoretical futures price using:

$$F = P[1 + t(r - c)].$$

- Explain whether the futures price should be above or below the cash price.
- If the market futures price is 103, identify the arbitrage trade and compute the approximate arbitrage profit.
- If the market futures price is 99, identify the arbitrage trade and compute the approximate arbitrage profit.
- Explain why arbitrage pressure pushes the futures price toward the theoretical price.

3.5 Question 15

A portfolio manager wants to hedge a long bond portfolio using Treasury futures. The portfolio has DV01 of \$27,840. The relevant futures contract has a cheapest-to-deliver bond with:

- modified duration = 6.5;
- CTD clean price = 98.50;
- deliverable face amount = \$100,000;
- conversion factor = 0.9200.

Answer the following:

- Explain why Treasury futures do not have a true Macaulay or modified duration.
- Compute the CTD bond's market value for the deliverable face amount.
- Compute the CTD bond's DV01.
- Compute the approximate futures DV01.
- Compute the number of futures contracts needed for a full DV01 hedge and state whether the manager should buy or sell futures.

4 Lecture 10: Credit Spread Exposure and Credit Risk Modeling

4.1 Question 16

A corporate bond has market value 4,000,000, spread duration of 4.5, and a credit spread of 180 basis points.

Answer the following:

- Explain what spread duration measures.
- Estimate the bond's percentage price change if its spread widens by 35 basis points.
- Estimate the dollar price change.
- Compute duration times spread (DTS).
- Explain why DTS can be more useful than spread duration alone when comparing bonds with different credit quality.

4.2 Question 17

A corporate bond has analytical duration of 6.0. A regression of its weekly return on the weekly Treasury return produces a slope coefficient of 0.65.

Answer the following:

- Distinguish analytical duration from empirical duration.
- Compute the bond's empirical duration using the duration-multiplier approach.
- Estimate the percentage price change associated with a 40-basis-point increase in Treasury yields using analytical duration.
- Repeat the estimate using empirical duration.
- Explain why the empirical estimate may be smaller for a lower-rated corporate bond.

4.3 Question 18

In a one-period structural credit model, a firm has asset value of \$120 million today. At the debt maturity date, it must repay debt with face value of \$100 million.

Answer the following:

- State the condition under which the firm defaults at maturity.
- Describe the bondholders' payoff at maturity.
- Describe the shareholders' payoff at maturity and explain its option interpretation.
- Explain how an increase in asset volatility affects default risk, holding other inputs constant.
- Give two practical limitations of structural credit models.

4.4 Question 19

Two portfolios each contain 100 equal-sized issuers. Every issuer has a one-year default probability of 2%. Defaults are independent in Portfolio A but strongly exposed to a common recession factor in Portfolio B.

Answer the following:

- Compute the expected number of defaults in each portfolio.
- Explain whether the expected number of defaults differs across the portfolios.
- Identify which portfolio has greater tail risk and explain why.
- Explain why pairwise correlation may not fully describe credit-portfolio risk.
- Explain what a copula adds to the marginal default distributions.

4.5 Question 20

A reduced-form model assumes a constant annual default intensity of $\lambda = 0.02$ and a loss given default of 60%.

Answer the following:

- a. Interpret the annual default intensity in words.
- b. Compute the one-year survival probability and default probability.
- c. Compute the five-year survival probability and cumulative default probability.
- d. For 100 similar issuers that survive to the start of the year, compute the expected number of defaults over the next year.
- e. Using spread $\approx \lambda \times \text{LGD}$, estimate the credit spread in basis points and state one reason why an observed market spread may differ.